

Guide to Writing Effective AI Prompts for R-Script Assistance

For Wageningen University & Research Students

Introduction

This guide helps you write effective prompts when using AI tools (like ChatGPT, Claude, Mistral, or others) to assist with R programming for your coursework, research projects, or data analysis tasks at WUR. Good prompts lead to better, more accurate, and more useful responses.

The Anatomy of an Effective Prompt for R-Script Help

1. Context Setting

What to include: - Your academic level (BSc, MSc, PhD) - The course or research project context - Your current R skill level (beginner, intermediate, advanced) - The broader goal of your analysis

Why it matters: The AI can adjust its explanations and code complexity to match your level and ensure the solution fits your academic context.

Example:

"I'm an MSc student in Plant Sciences at WUR working on my thesis. I have intermediate R skills and am analyzing field trial data for crop yield responses to different fertilizer treatments."

2. Specific Problem Description

What to include: - What you're trying to accomplish - What you've already tried (if anything) - Where you're stuck - Any error messages you're encountering (copy them exactly)

Why it matters: Clarity about your specific problem helps the AI provide targeted solutions rather than generic advice.

Example:

"I need to create a linear mixed model to analyze my data, but I'm getting this error: 'Error in lmer(yield ~ treatment + (1|block)): could not find function lmer'. I've tried installing lme4 but it didn't work."

3. Data Structure Information

What to include: - Type of data (continuous, categorical, count data, etc.)
- Data structure (data frame, list, matrix, etc.) - Number of observations and variables - Key variable names and what they represent - Any hierarchical or nested structure

Why it matters: R code is highly dependent on data structure. Without this information, the AI might suggest code that doesn't work with your specific data.

Example:

```
"My data frame has 240 observations with these columns:  
- plot_id (factor, 1-240)  
- block (factor, 1-4)  
- treatment (factor, 3 levels: control, low, high)  
- yield_kg (numeric, continuous)  
- soil_type (factor, 2 levels: sandy, clay)"
```

4. Desired Output or Goal

What to include: - What you want the code to produce - Preferred format (plot, table, statistical output, etc.) - Any specific requirements or constraints - Whether you need publication-quality outputs

Why it matters: Being specific about the end goal ensures you get code that produces exactly what you need.

Example:

```
"I want to create a boxplot showing yield distribution for each  
treatment, with different colors for each treatment, and error bars  
showing standard error. The plot should be publication-ready with  
proper axis labels and a legend."
```

5. Constraints and Requirements

What to include: - Required or preferred R packages (e.g., tidyverse, base R) - Packages you cannot use (due to installation issues or course requirements) - Coding style preferences (pipes vs. base R) - Any specific methodological requirements from your supervisor or course

Why it matters: This prevents the AI from suggesting solutions you can't implement.

Example:

"Please use tidyverse syntax as that's what our course teaches.
I cannot install any packages that aren't on CRAN. My supervisor
requires that I use type III sums of squares for the ANOVA."

How to Structure Your Initial Prompt

Template for First-Time Questions:

[CONTEXT]

I am a [your level] student at WUR studying [your program].
I have [beginner/intermediate/advanced] R skills.

[SPECIFIC PROBLEM]

I'm trying to [specific task], but [specific problem/error].
[Include error message if applicable]

[DATA STRUCTURE]

My data looks like this:

- Variable 1: [type and description]
 - Variable 2: [type and description]
- [Include sample data if possible using dput() output]

[DESIRED OUTPUT]

I need to produce [specific result/visualization/analysis].

[CONSTRAINTS]

- Please use [specific packages/approach]
 - I cannot use [restricted items]
 - Requirements: [any specific requirements]
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How to Update or Refine Your Prompt (Iterative Prompting)

When the First Response Doesn't Fully Solve Your Problem:

1. Acknowledge what worked:

"Thanks! The code worked for creating the plot, but..."

2. Specify what needs adjustment:

"...I need to modify it so that:

- The x-axis labels are rotated 45 degrees
- The legend appears at the bottom

- Each treatment group shows individual data points overlaid on the boxplot"

3. Provide error messages if new ones appear:

"When I tried to add the theme elements you suggested, I got this error:
[paste exact error]"

When You Want to Start Fresh:

If the conversation has gotten confused or gone off-track:

"Let me restart and clarify what I need:
[Provide a fresh, complete description using the template above]"

What to Check in the AI's Response

1. Code Correctness

- ☐ Does the code run without errors?
- ☐ Are all required packages loaded?
- ☐ Are variable names consistent with your data?
- ☐ Are data types handled correctly?

2. Statistical Appropriateness

- ☐ Is the statistical method appropriate for your data type?
- ☐ Are assumptions of the test met?
- ☐ Is the model specification correct (e.g., random effects properly specified)?
- ☐ Are multiple testing corrections applied if needed?

3. Code Understanding

- ☐ Do you understand what each line does?
- ☐ Can you explain the logic to your supervisor or peers?
- ☐ Are there comments explaining complex sections?
- ☐ Could you modify this code for similar future tasks?

4. Output Quality

- ☐ Does the output answer your specific question?
- ☐ Are visualizations clear and properly labeled?
- ☐ Are results formatted appropriately?
- ☐ Do the results make biological/scientific sense?

5. Best Practices

- ☐ Is the code efficient (not unnecessarily complex)?
- ☐ Are there any deprecated functions?

- ☐ Is the code well-organized and readable?
 - ☐ Are there warnings you should be aware of?
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Common Scenarios at WUR and Example Prompts

Scenario 1: Statistical Analysis for Field Trials

I'm a Plant Sciences MSc student analyzing a field trial with a randomized complete block design. I have 4 blocks and 6 variety treatments. My response variable is grain yield (kg/plot).

Data structure:

- block: factor with 4 levels
- variety: factor with 6 levels
- yield: numeric (continuous)
- 24 total observations

I need to:

1. Perform an ANOVA with block as a random effect
2. Check model assumptions (normality, homogeneity)
3. If significant, conduct post-hoc pairwise comparisons with Tukey adjustment
4. Create a bar plot showing mean yields with error bars (SE)

Please use lme4 for mixed models and emmeans for post-hoc tests.

Scenario 2: Data Cleaning and Preparation

I'm working with weather data for my thesis on crop modeling. I have a CSV file with daily weather measurements but there are several issues:

Data problems:

- Some temperature values are -999 (missing data code)
- Dates are in format DD-MM-YYYY but need YYYY-MM-DD
- Column names have spaces and special characters
- Some duplicate date entries

I need R code to:

1. Read the CSV and rename columns to snake_case
2. Convert date format
3. Replace -999 with NA
4. Remove duplicate dates (keeping first occurrence)
5. Create a summary showing how many missing values per variable

Please use tidyverse functions (readr and dplyr).

Scenario 3: Creating Publication-Quality Figures

I need to create a figure for my thesis showing how soil organic carbon (SOC) changes across three land use types over time.

My data:

- land_use: factor (forest, agriculture, grassland)
- year: numeric (2010-2023)
- SOC: numeric (% carbon)
- site_id: factor (5 sites per land use)

Requirements:

1. Line plot with separate lines for each land use type
2. Points showing actual measurements
3. Smooth trend lines (loess)
4. WUR color palette if available (otherwise colorblind-friendly)
5. Proper axis labels with units
6. Legend with clear names
7. Size 6x4 inches, 300 dpi for publication

Please use ggplot2 and provide both the code and theme settings.

Scenario 4: Debugging Code

I'm trying to calculate summary statistics by group but my code isn't working. Here's what I have:

[paste your code here]

And I'm getting this error:

[paste exact error message]

My data frame 'field_data' has:

- treatment: factor with 3 levels
- replicate: numeric 1-4
- biomass: numeric values

I want mean and standard deviation of biomass for each treatment. What am I doing wrong?

Tips for Better Results

Do's:

Provide sample data using `dput(head(your_data))` when possible. Copy exact error messages including any warnings. Mention if you're following a specific textbook or method. Ask for explanations if you don't understand something. Request alternatives if one approach doesn't work. Specify if you need code comments for learning purposes.

Don'ts:

Don't ask "How do I do everything in R?" - be specific. Don't assume the AI knows your previous context without reminding it. Don't paste huge amounts of code without explaining what it should do. Don't use the AI-generated code without understanding it. Don't ignore warnings just because code runs. Don't forget to cite AI assistance in your work per WUR guidelines.

Verifying and Learning from AI Responses

Always Cross-Check:

1. **Test with your actual data:** Don't just assume the code will work.
2. **Consult R documentation:** Use `?function_name` to understand what functions do.
3. **Check methodology:** Verify statistical approaches with textbooks or papers.
4. **Ask your supervisor:** Show them the approach for validation.
5. **Compare with course materials:** Ensure consistency with what you've learned.

Build Your Understanding:

- Ask "Why did you use this function instead of X?"
 - Request explanations: "Can you explain what this line does?"
 - Ask for alternatives: "Is there another way to accomplish this?"
 - Inquire about best practices: "Is this the most efficient approach?"
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Example Iterative Conversation

First Prompt:

I'm a BSc student in Environmental Sciences. I have a dataset of water quality measurements from different locations along a river. I need help creating a map showing sampling points colored by

pollution level. I have coordinates (lat/long) and pollution scores (1-10). How do I do this in R?

After First Response:

Thanks! The basic plot works, but I need to:

1. Add a basemap showing the actual river
2. Make the points sized by pollution level (not just colored)
3. Add a scale bar and north arrow
4. Export it as a high-resolution PNG

Can you help me modify the code?

Further Refinement:

The basemap code gives an error about API key. Can you show me how to do this with a different package that doesn't need an API key? I've heard about the 'sf' and 'ggmap' packages.

Ethical Use and Academic Integrity at WUR

Proper Use:

- Use AI to learn and understand R concepts
- Use AI to debug and improve your code
- Use AI to explore different approaches to problems
- Document AI assistance in your work

Improper Use:

- Submitting AI-generated code as entirely your own work without understanding
- Using AI to complete entire assignments without learning
- Not citing AI assistance where required
- Bypassing learning objectives of your course

WUR Guidelines:

Always check your course manual or ask your instructor about specific rules regarding AI assistance. Different courses may have different policies.

General principle: You should be able to explain every line of code you submit and understand why it works.

Troubleshooting Common Issues

“The code doesn’t work with my data”

→ Provide actual data structure using `str(your_data)` or sample rows → Check if variable names match exactly (R is case-sensitive)

“I get an error about missing packages”

→ Tell the AI which packages you have/can install → Ask for alternatives using base R or different packages

“The output isn’t what I expected”

→ Describe specifically what you got vs. what you wanted → Share a screenshot or description of the unexpected output

“The code is too complex for me to understand”

→ Ask for a simpler version with more comments → Request step-by-step explanations → Ask for the code to be broken into smaller pieces

Additional Resources

WUR-Specific:

- WUR Library R workshops and materials
- Your course documentation and required textbooks
- R user groups at WUR (check with your study program)

General R Resources:

- R Documentation: `?function_name` or `help(package="packagename")`
- Stack Overflow (for error messages and specific problems)
- R-bloggers for tutorials and examples
- Posit Community for tidyverse-related questions

Statistical Consultation:

For complex statistical questions, consider: - Mathematical and Statistical Methods (MSM) group at WUR - Your thesis supervisor or course instructor - WUR Statistics support services

Final Checklist Before Submitting Your Work

- ☐ I understand every line of code I'm submitting
 - ☐ I've tested the code thoroughly with my actual data
 - ☐ I've verified the statistical approach is appropriate
 - ☐ I've checked my results make biological/scientific sense
 - ☐ I've properly cited any AI assistance according to course guidelines
 - ☐ I've added comments to my code explaining key steps
 - ☐ I can explain my methodology to my supervisor
 - ☐ I've saved my code with proper version control
 - ☐ My visualizations are clear and publication-ready (if required)
 - ☐ I've documented any packages and versions used
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Questions or Problems?

If you're still stuck after trying these prompting strategies: 1. Book a consultation with WUR's data support services 2. Attend R help sessions offered by your department 3. Form a study group with peers to work through problems together 4. Consult with your course instructor during office hours

Remember: AI is a powerful learning tool, but it's not a replacement for understanding the concepts and methods you're using in your research.

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